

THE PETRIDISH



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Welcome to the second edition of our newsletter !

The PetriDish, the official newsletter of BioSoc-DTU at Delhi Technological University (formerly DCE) serves as a vibrant hub for the latest advancements and insights in the world of biotechnology and acts as a bridge connecting students, researchers, industry professionals, and biotech enthusiasts, with its vision to keep you informed and inspired with comprehensive articles, news updates, and event highlights from BioSoc-DTU, the Department of Biotechnology at DTU, and the global biotech community.

The second edition of The PetriDish features exciting updates from the world of biotechnology, including recent news and breakthroughs, as well as highlights from BioSoc's latest events. It also includes a fascinating article on brain wave noise cancellation, showcasing cutting-edge research and its potential applications.

BIOBYTES

Fish guts might play an important role in future skin care products

Researchers might have found something even weirder than snail slime to put on your face: molecules made by fish gut bacteria. In cultured cells, the compounds had skin-brightening and anti-wrinkle properties, making them potential ingredients for your future skin care routine.

Miniature treadmills accelerate studies of insects walking

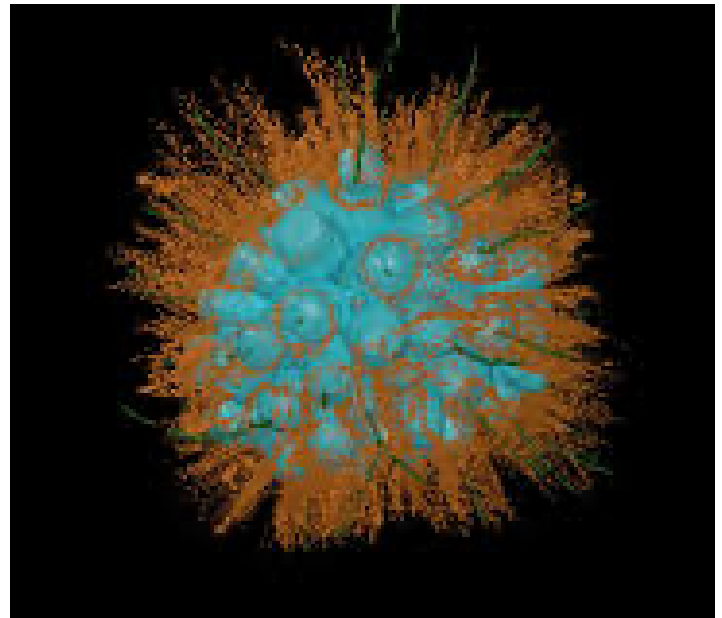
Fruit flies walking on miniature treadmills are helping scientists learn how the nervous system enables animals to move in an unpredictable and complex world. The researchers engineered these small-scale machines from inexpensive parts.

India aims to lead next industrial revolution with biomanufacturing policy: DBT

India introduced the BioE3 policy to tackle challenges related to food, energy, and climate. Approved by the Union Cabinet, the policy focuses on enhancing innovation in biomanufacturing, covering everything from medicines to bioplastics. The initiative is expected to create job opportunities and promote sustainable development.



India's BioE3 policy addresses food, energy, and climate challenges by promoting biomanufacturing innovations such as medicines and bioplastics. Approved by the Union Cabinet, it aims to generate jobs and support sustainable development.



Mono Lake, with its salty, arsenic-laden water, houses brine shrimp, alkali flies, and a newly discovered microscopic choanoflagellate. This organism forms colonies with unique bacterial microbiomes and is related to the ancestors of all animals, potentially offering insights into animal-bacteria interactions and the evolution of multicellular life.

Engineers design lookalike drug carrier to evade lung's lines of defense

Managing hard-to-treat respiratory illnesses like asthma and pulmonary fibrosis just got easier if a new drug-carrying molecule is as sneaky as its inventors think.

Creature the size of a dust grain found hiding in California's Mono Lake

Mono Lake is a beautiful but harsh environment, its salty and arsenic-laced water home to brine shrimp, alkali flies and little else. Scientists recently discovered an unsuspected resident, however, a microscopic creature -- a choanoflagellate -- that forms colonies that harbor their own unique bacterial microbiomes. The creature, part of the sister group to all animals, could shed light on the evolution of animals' intimate interactions with bacteria and the rise of multicellular life.

~Hunar Aggarwal

SCIENTIST IN SPOTLIGHT

Richard Mooney is a renowned neurobiologist and professor at Duke University. He holds the George Barth Geller Professorship for Research in Neurobiology and is a key figure at the Duke Institute for Brain Sciences. Mooney's research focuses on the neural circuits involved in learning and perception, with particular emphasis on vocal learning in birds and humans, making significant contributions to our understanding of how the brain processes and produces learned behaviours such as speech

Richard Mooney, a neurobiologist at Duke University, is a key scientist studying the brain's natural noise-cancelling mechanisms. His work focuses on how the brain suppresses self-generated sounds, allowing organisms to better detect external, potentially dangerous noises like a predator's approach. Mooney's research shows that the motor cortex sends signals to the auditory cortex to filter out predictable noises, like footsteps, allowing for better attention to external sounds. His experiments with mice have revealed that this process happens in real-time and can be learned over time.



Mooney's groundbreaking research offers insights into how the brain differentiates between self-made and external sounds, providing a biological basis for noise cancellation that mirrors technology like noise-cancelling headphones. His studies are crucial for understanding sensory processing and motor coordination in the brain, and his work has been supported by several prestigious organizations, including the Howard Hughes Medical Institute and the National Institutes of Health

The research aims to uncover the neural mechanisms through which experience shapes learning, behavior, and perception. We focus on sensorimotor circuits involved in vocal communication in songbirds and auditory-motor integration in mice. Using advanced techniques like multiphoton imaging, neural recording, optogenetics, and viral transgenic methods, we investigate circuits critical to vocal learning and auditory perception.

~Samihan Sharma

EVENT IN SPOTLIGHT

BioSoc-DTU organized its annual orientation on September 13, 2024, intending to introduce and familiarize the juniors of this academic year to the Department of Biotechnology, DTU and Biosoc itself and to foster an interest in Life Sciences. The event was sponsored by Hurricane Energy Drink, Bistro 57, Nescafe. The Emcee then proceeded to express gratitude on the society's behalf to Prof. Yasha Hasija, Dr Navneeta Bhardvaja and Dr Prakash Chandra for their indispensable support for the society.

The Speaker session began with Dr VTS Pavan Kumar Kavaluru - VP, R&D at the esteemed "Catalysts Group Pvt Ltd", a leading biotechnology firm providing innovative and disruptive enzyme-based solutions to industries across India and other developing countries. Dr Kavaluru in his session comprehensively explained the workings of his company, the various industries they serve with their enzyme-based solutions and the immense savings offered by their products to these companies. Dr Kavaluru in particular focused on three domains that Catalysts Group Specializes in: sugarcane processing, Distilling & Brewing and Biogas production, explaining these topics in detail.



No sooner, Dr Kavaluru was facilitated by respected faculty members, thanking him for his informative presentation and presence.

At that point, the society's introduction proceeded with a brief about BioSoc-DTU's mission and Vision and how the society is affiliated with the renowned Royal Society of Biotechnology. A short slideshow of the past events was then given to showcase the participation of the society. The achievements of notable alumni were recounted, many of whom are GATE top scorers and some are holding jobs and positions in esteemed institutions and companies. Following this, the co-heads of the society's different departments and divisions described the function and ambitions of their respective departments.

The first edition of "The Petridish" was launched with the ideology of the newsletter explained by the respective co-editors. Lastly, the sponsors were thanked again for their support, followed by a vote of thanks, expressing gratitude towards the esteemed speaker, faculty members and the audience marking the end of a successful orientation.

~Mahim Kamble

FEATURE FOCUS

Brainwave Noise Cancellation: A Game-Changer for Better Sleep

Neuromodulation is becoming the up-and-coming approach that could help improve sleep by targeting and suppressing alpha waves related to brain waves associated with relaxed wakefulness. Neuromodulation enhances deeper and more restorative sleep by reducing such waves.

One of the most exciting innovations in this area comes from a Cambridge, Massachusetts-based startup, Elemind. This startup has created a wearable device worth \$349, designed to improve sleep through neuromodulation. According to co-founder and CEO Meredith Perry, the device tracks the electrical activity of the brain via EEG sensors and then modulates brainwave patterns through the use of sound delivered via bone conduction.

What is innovative about Elemind's headband is the non-intrusive approach. While other devices use biofeedback, which requires active participation by the users in self-regulating their brainwaves, Elemind's device is the first to try directly influencing brain activity with auditory stimuli, not requiring any effort by the wearer. Pink noise, which is a hissing sound similar to white noise but with lower, higher-frequency components, is played via bone conduction through the headband, allowing it to impact brain activity without interfering with the sleeping environment of the user.

Clinical trials indicated that 76% of users experienced falling asleep faster with the headband and, on average, reduced latency to sleep onset by 48%.



Perry says it is like "noise cancellation for the brain." The mental chattering that interferes with sleep slowly comes to an end. This lightweight, comfortable device houses multiple EEG sensors, a processor, an accelerometer, and a rechargeable battery inside its cushioned band that rests across the forehead.

Elemind's development in sleep technology is a radical transformation. Being the first device to use auditory stimuli for passive brain function regulation, it provides the conceptual groundwork for an innovative intervention approach to sleep. This development may signal the start of the next stage in sleep interventions to potentially replace traditional sleep aids, as neuromodulation becomes an increasing focus. Better sleep technology has finally arrived, and it promises to be more natural and effective at resolving restorative sleep.

~Unnati Nath

OUR TEAM

EDITORS

Apurva

M. Sc Biotechnology, 2nd year

Shashwat Jha

B.Tech Electrical Engineering, 2nd year

CONTENT

Hunar Aggarwal

B.Tech Biotechnology, 2nd year

Mahim Kamble

B.Tech Biotechnology, 2nd year

Samihan Sharma

B.Tech Biotechnology, 2nd year

Unnati Nath

B.Tech ECE, 3rd year

DESIGN TEAM

Krishna Singhal

B.Tech Biotechnology, 2nd Year

Saksham Gupta

B.Tech Biotechnology, 2nd Year



@biosocdtu



@BioSoc-DTU

